Compound angle equations

1 Consider the equation $\cos(\theta + 20^\circ) = \frac{1}{\sqrt{2}}$ for $0^\circ \leq \theta \leq 360^\circ$.

a If $\alpha = \theta + 20^\circ$, find the solutions of $\cos\alpha = \frac{1}{\sqrt{2}}$ for $20^\circ \leq \alpha \leq 380^\circ$.

b Hence, solve $\cos(\theta + 20^\circ) = \frac{1}{\sqrt{2}}$.

2 Consider the equation $\cos(\theta - 100^\circ) = \frac{1}{2}$ for $0^\circ \leq \theta \leq 360^\circ$.

a If $\alpha = \theta - 100^\circ$, find the solutions of $\cos\alpha = \frac{1}{2}$ for $-100^\circ \leq \alpha \leq 260^\circ$.

b Hence, solve $\cos(\theta - 100^\circ) = \frac{1}{2}$.

3 Consider the equation $\sin(\theta + 35^\circ) = \frac{1}{2}$ for $0^\circ \leq \theta \leq 360^\circ$.

a If $\alpha = \theta + 35^\circ$, find the solutions of $\sin\alpha = \frac{1}{2}$ for $35^\circ \leq \alpha \leq 395^\circ$.

b Hence, solve $\sin(\theta + 35^\circ) = \frac{1}{2}$.

4 Consider the equation $\sin(\theta + 25^\circ) = \frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta \leq 360^\circ$.

a If $\alpha = \theta + 25^\circ$, find the solutions of $\sin\alpha = \frac{\sqrt{3}}{2}$ for $25^\circ \leq \alpha \leq 385^\circ$.

b Hence, solve $\sin(\theta + 25^\circ) = \frac{\sqrt{3}}{2}$.

5 Consider the equation $\tan(\theta + 15^\circ) = \frac{1}{\sqrt{3}}$ for $0^\circ \leq \theta \leq 360^\circ$.

a If $\alpha = \theta + 15^\circ$, find the solutions of $\tan\alpha = \frac{1}{\sqrt{3}}$ for $15^\circ \leq \alpha \leq 375^\circ$.

b Hence, solve $\tan(\theta + 15^\circ) = \frac{1}{\sqrt{3}}$.

6 Solve the following equations for $0^\circ \leq \theta \leq 360^\circ$:

a $\cos(\theta + 60^\circ) = 1$

b $\cos(\theta - 75^\circ) = -\frac{1}{\sqrt{2}}$

c $\sin(\theta + 45^\circ) = -\frac{\sqrt{3}}{2}$

d $\tan(\theta - 45^\circ) = \frac{1}{\sqrt{3}}$

e

f $2\sin(\theta - 60^\circ) = -\sqrt{3}$

7 Solve the following equations for $0^\circ \leq \theta \leq 360^\circ$. Round your answers to two decimal places if necessary.

a $\cos 3\theta = \dfrac{1}{\sqrt{2}}$

b $\sin 2\theta = -\dfrac{1}{2}$

c $\cos 2\theta = 0.7$

d $2 \sin 3\theta - \sqrt{2} = 0$

e $\sin\left(\dfrac{\theta}{2}\right) = \dfrac{\sqrt{3}}{2}$

f $\tan 3\theta = -1$

8 Solve the following equations for $0^\circ \leq x \leq 180^\circ$:

a $\tan 4x = \sqrt{3}$

b $\sin 2x = \dfrac{\sqrt{3}}{2}$

c $\cos 3x = -\dfrac{1}{\sqrt{2}}$

d $\tan\left(\dfrac{x}{3}\right) = 1$

9 Solve the following equations for $-180^\circ \leq x \leq 180^\circ$:

a $\cos 3x = -\dfrac{1}{\sqrt{2}}$

b $\sin 2x = \dfrac{1}{2}$

c $\tan\left(\dfrac{x}{2}\right) = \pm\sqrt{3}$

d $\sin 3x - 1 = 0$

Squared equations

10 Solve the following equations for $0^\circ \leq \theta \leq 360^\circ$. Round your answers to two decimal places if necessary.

a $2 \sin^2 \theta = 1$

b $\sin^2 \theta = \dfrac{3}{4}$

c $\cos^2 \theta = \dfrac{1}{2}$

d $\tan^2 \theta = \dfrac{1}{3}$

e $4 \sin^2 \theta = 1$

f $\cos^2\left(\dfrac{\theta}{2}\right) - 1 = 0$

g $\sin^2 2\theta = \dfrac{1}{2}$

h $\cos^2 3\theta = \dfrac{3}{4}$

11 Solve the following equations for the given domain:

a $\cos^2 \theta = \dfrac{3}{4}$ for $0^\circ < \theta < 90^\circ$

b $4 \sin^2 \theta = 3$ for $90^\circ \leq \theta \leq 270^\circ$

Factorisation

12 State the number of solutions for θ of the equation $\left(\sin\theta+\frac{\sqrt{3}}{2}\right)\left(\cos\theta+\frac{\sqrt{3}}{2}\right)=0$ for $0^\circ<\theta<90^\circ$.

13 Solve the following equations for $0^\circ\leq\theta\leq360^\circ$. Round your answer to one decimal place if necessary.

- **a** $\tan^2(\theta)-2\tan\theta+1=0$
- **b** $4\cos^3(\theta)=3\cos\theta$
- **c** $5\sin^2(\theta)+8\sin\theta\cos\theta-4\cos^2(\theta)=0$
- **d** $7\cos^2(\theta)+3\cos\theta=3$
- **e** $9\sin^2(x)+5\sin x=3$
- **f** $\cos\theta\tan\theta=\cos\theta$
- **g** $8\tan^2(\theta)\cos\theta-4\tan^2(\theta)=0$

14 Solve the following equations for $0^\circ<\theta<90^\circ$:

- **a** $\left(\sin\theta-\frac{\sqrt{3}}{2}\right)\left(\cos\theta-\frac{1}{\sqrt{2}}\right)=0$
- **b** $\left(\sin\theta+\frac{1}{\sqrt{2}}\right)\left(\tan\theta-\frac{1}{\sqrt{3}}\right)=0$

15 Solve the following equations for the given domain:

- **a** $\left(\tan\theta-\frac{1}{\sqrt{3}}\right)\left(\cos\theta+\frac{1}{\sqrt{2}}\right)=0$ for $180^\circ\leq\theta\leq270^\circ$
- **b** $\left(\sin\theta-\frac{1}{2}\right)\left(\cos\theta-\frac{1}{\sqrt{2}}\right)=0$ for $270^\circ\leq\theta\leq360^\circ$
- **c** $\tan^2(\theta)=\sqrt{3}\tan\theta$ for $-90^\circ<\theta<90^\circ$
- **d** $2\sin^2(\theta)-\sin\theta=0$ for $-180^\circ\leq\theta\leq180^\circ$

16 Deborah is solving the equation $2\sin^2\theta+7\sin\theta+5=0$. After some factorisation, she arrives at the pair of equations $\sin\theta+1=0$ and $2\sin\theta+5=0$. Which of the two equations has a solution?

Equations and trigonometric identities

17 Solve the equation $3\sin\theta = 3\sqrt{3}\cos\theta$ for $0 < \theta < 90$.

18 Solve the following equations for $0^{\circ} \leq \theta < 360^{\circ}$. Round your answers to the nearest degree if necessary.

a $2\cos^2(\theta) = 2 - \sin\theta$

b $\sin^5(\theta)\cos^3(\theta) = 0$

c $\cos^2(\theta) - 8\sin\theta\cos\theta + 3 = 0$

d $\dfrac{1-\tan^2(\theta)}{1+\tan^2(\theta)} + \cos\theta = 0$